



Semester Project

Analyzing measurements of sublimation and snow transport in Antarctica

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1 Introduction

Sublimation and deposition at the snow-atmosphere interface are critical processes affecting the surface mass balance of ice sheets. These phase changes are strongly influenced by environmental variables such as wind speed, humidity, and temperature. Snow transport, driven by strong winds, plays a central role in modulating these processes by increasing the exposure of snow grains to the atmosphere, altering their surface area and sublimation dynamics. Understanding this interplay is essential for accurately quantifying snow mass fluxes and the associated latent heat exchange. This project aims to investigate the impact of snow transport on sublimation and deposition rates using data collected at the Princess Elisabeth Antarctica station. The study involves applying and comparing different measurement techniques to analyze the relationships between wind-driven snow transport and water vapor fluxes. By characterizing these interactions, the project seeks to improve the understanding of snow mass fluxes in East Antarctica and their implications for the surface mass and energy balance.

2 Study area and experimental setup

The Princess Elisabeth Antarctica (PEA) station is a Belgian polar research facility located at 71°57′00″S, 23°20′51″E, at an altitude of 1300 m. The experimental setup is situated east of the station at 71.94958°S, 23.35013°E.

The data used in this project covers a full year, recorded from 29 November 2022 to 30 November 2023.



(a) PEA station location



(b) Sensor positions

Figure 1: Location of PEA station and sensor setup.

The following figure depicts the environmental monitoring station:

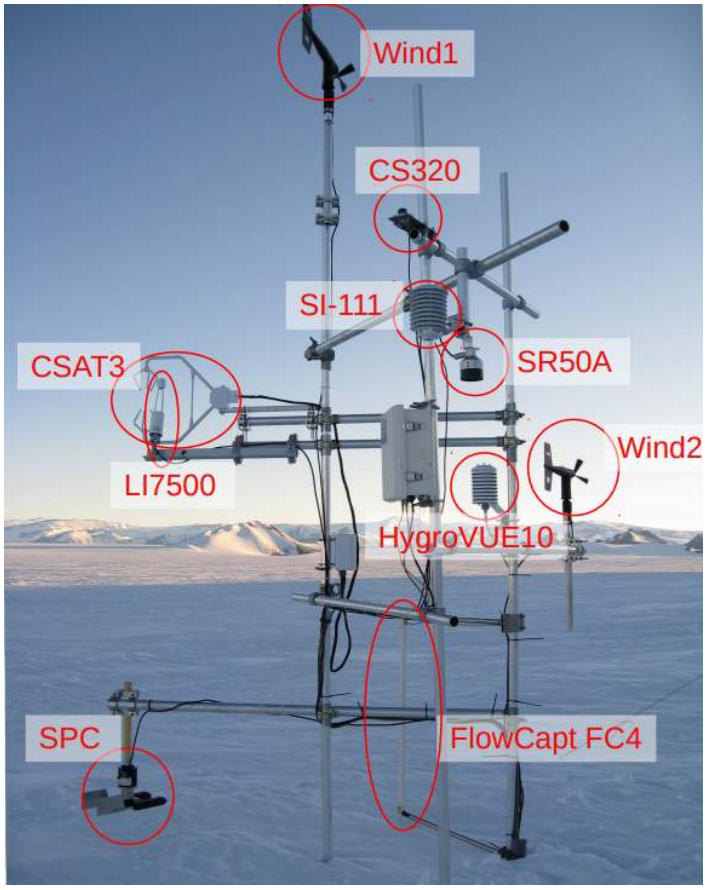


Figure 2: Caption

The sensors used in this project are :

Instrument	**Description**
CSAT3	Ultrasonic anemometer
FlowCapt FC4	Snow flux sensor
HygroVUE10	Temperature and relative humidity sensor
LI7500	Infrared gas analyzer (H_2O, CO_2)
SI-111	Infrared radiometer
SPC	SPC-95 Snow Particle Counter
SR50A	Sonic ranging sensor
Wind1, Wind2	Young wind monitor 05103

Table 1: List of used instruments

3 Data preprocessing

3.1 Data gaps and interpolation

The slow data, sampled at one-minute intervals, includes variables such as wind velocity, relative humidity, air temperature, surface temperature, snow height, and atmospheric pressure. Throughout the year, there were only a few data gaps observed, and these were linearly interpolated to maintain data continuity.

Gap duration (hh:mm:ss)	Occurrences
00:02:00	1
00:53:00	1
00:01:00	1
00:03:00	1

Table 2: Durations of data gaps in slow data.

3.2 Snow transport

The snow flux was measured using two different instruments:

- **SPC-95**: This instrument operates by detecting interferences in a laser beam caused by snow particles. It can determine the particle size, classifying it into one of 64 bins ranging from 36 μm to 500 μm.
- **FlowCapt FC4**: This instrument consists of a vertical tube that estimates the snow flux by analyzing the sound waves generated by snow particles colliding against it.

These two instruments provide measurements that are not expected to be identical. The SPC-95 measures snow flux over a small rectangular area ($25 \times 2mm$), while the FlowCapt FC4 computes an average flux over a vertical rectangular region ($32 \times 920mm$). Given that the flux is height dependent, the snow flux measured by each instrument inherently differs.

In the case of the SPC-95 sensor, recorded values were only available up to the end of April, and the raw data had not been temperature-corrected. The FlowCapt FC4 data exhibited occasional spikes. The objective of this analysis was to compare the temperature-corrected SPC-95 data with the FC4 data to determine whether the observed spikes were real and to assess the degree of correlation between the measurements from the two sensors.

3.2.1 Temperature correction

The SPC-95 sensor is known to underestimate particle size as temperature increases. Calibration was performed using the manufacturer's datasheet, which provides measurements for five different particle sizes at six distinct temperatures. This procedure ensures that the sensor's readings can be appropriately adjusted for temperature-related biases.

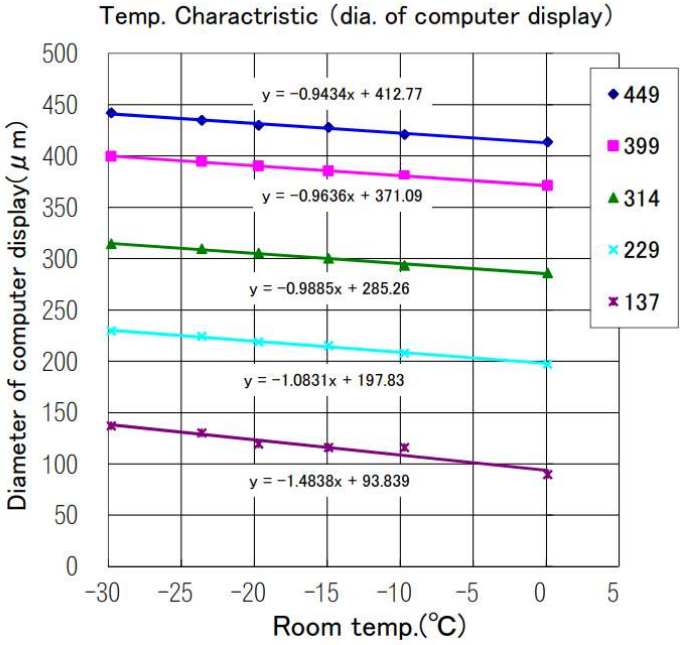


Figure 3: Calibration data for the SPC-95 sensor.

A custom calibration script was developed to adjust the sensor's data points. The script identifies the closest correction factor for each measurement (based on temperature and measured diameter) and interpolates the value to the reference calibration temperature of $-30°C$.

As shown in Figure 4, the corrected snow flux is slightly higher than the raw flux, which is reasonable given that January in the study region corresponds to summer and the temperature is relatively high.

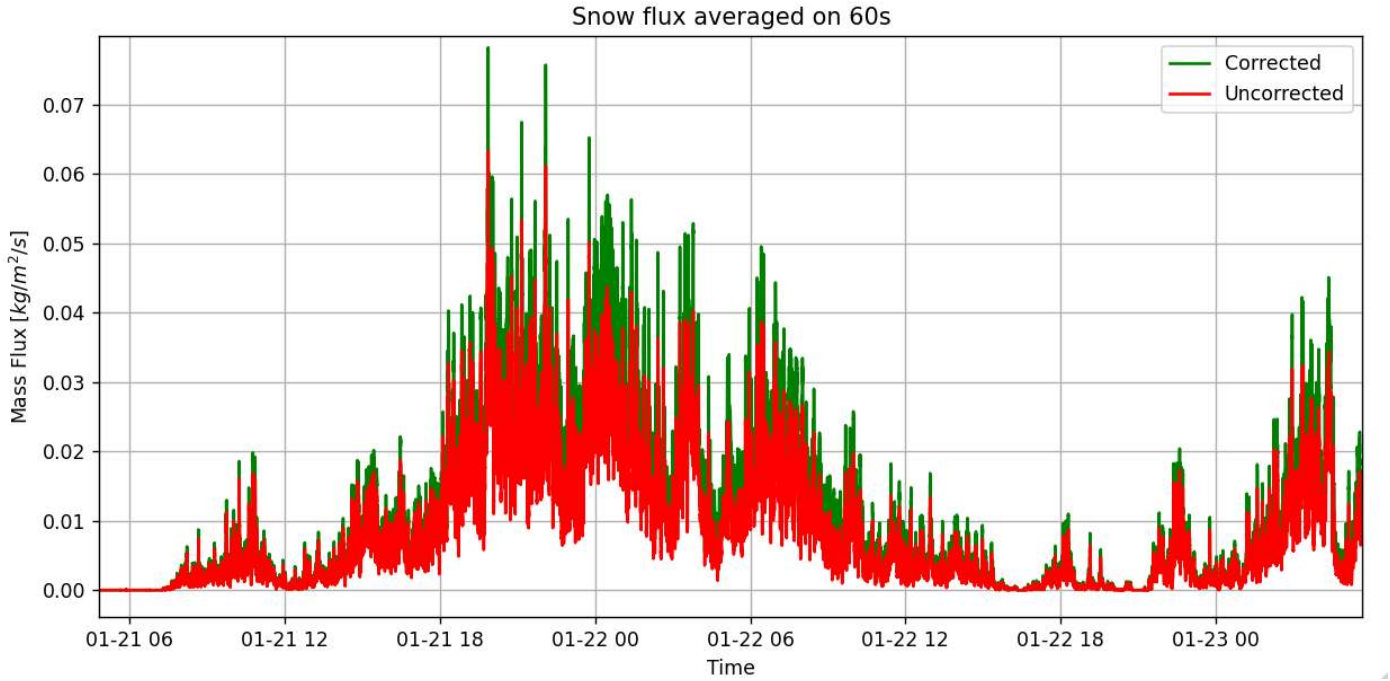


Figure 4: Comparison between corrected and raw snow flux.

Figure 5 presents the histogram of particle size distributions. There is a bump in the largest bin, suggesting that particles may be larger than 500 μm, indicating a continued underestimation of the actual snow flux. The bump around 180 μm in the corrected distribution occurs when two raw bins, adjusted with different slopes, end up rounded into the same corrected bin.

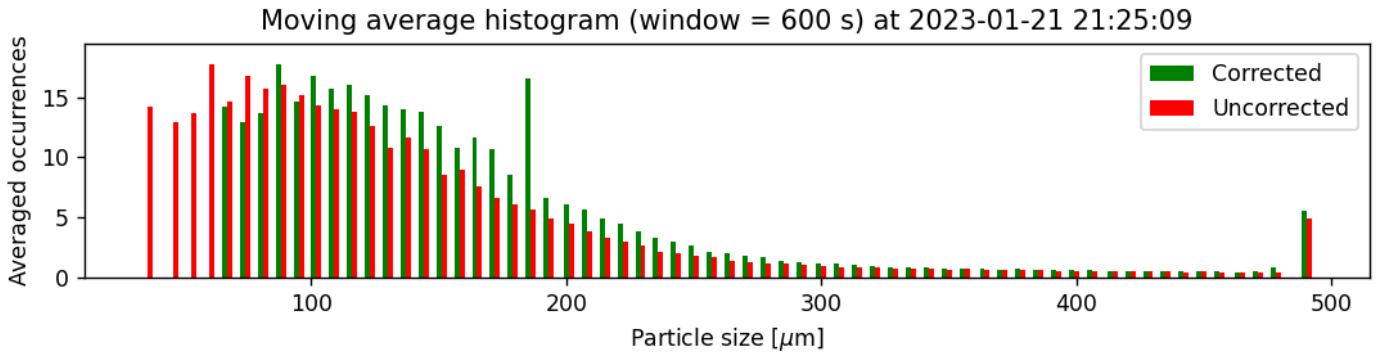


Figure 5: Histogram of corrected and raw particle size distribution.

3.2.2 Comparison and cleaning

As illustrated in Figures 6 and 7, the two sensors exhibit a good correlation, with the FC4 sensor generally recording higher values. However, in some instances, the FC4 data display suspicious spikes without corresponding increases in the SPC-95 measurements or the wind speed. These anomalies, which occurred only about ten times throughout the entire year, were manually removed from the dataset.

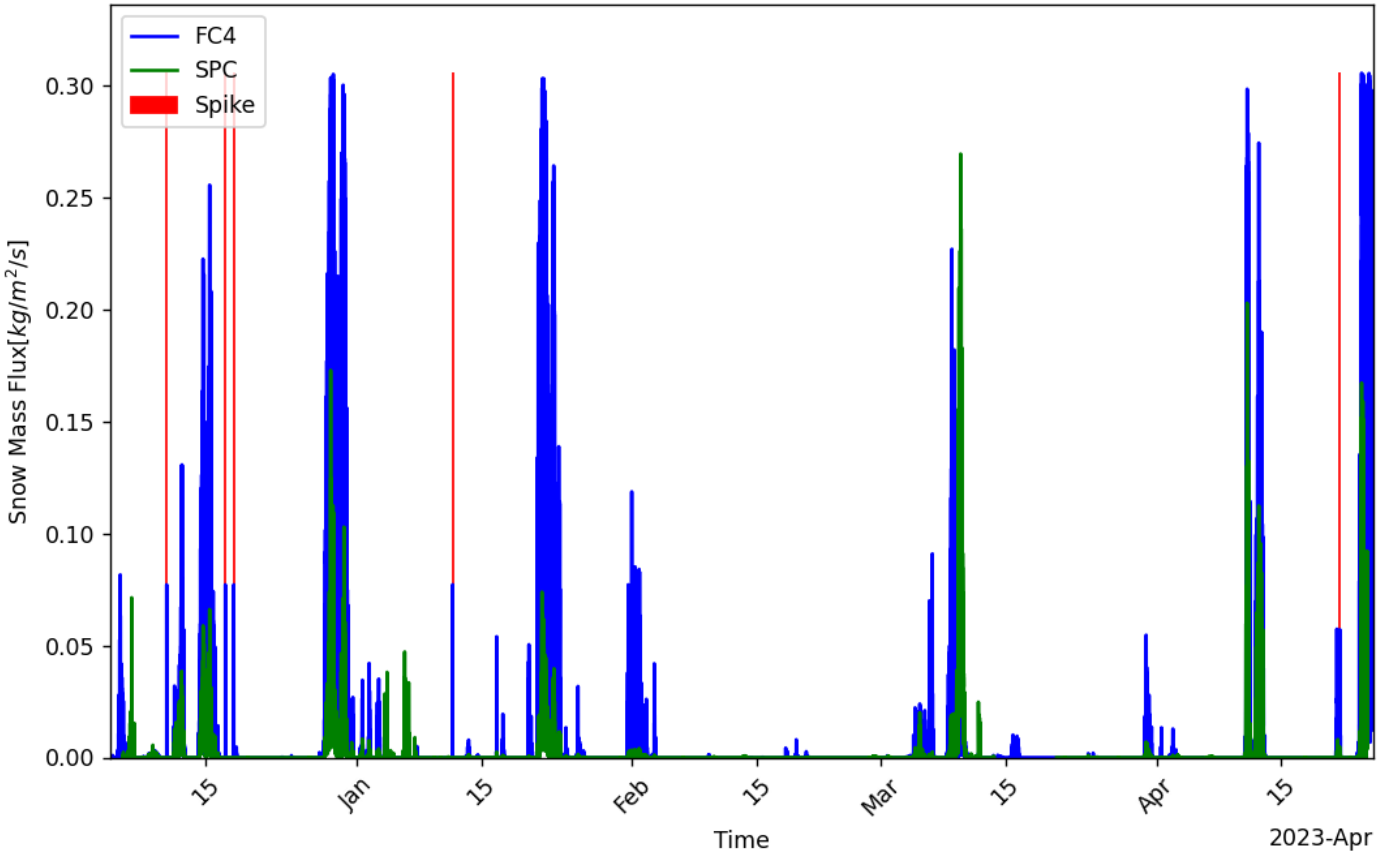


Figure 6: Comparison over the entire measurement period.

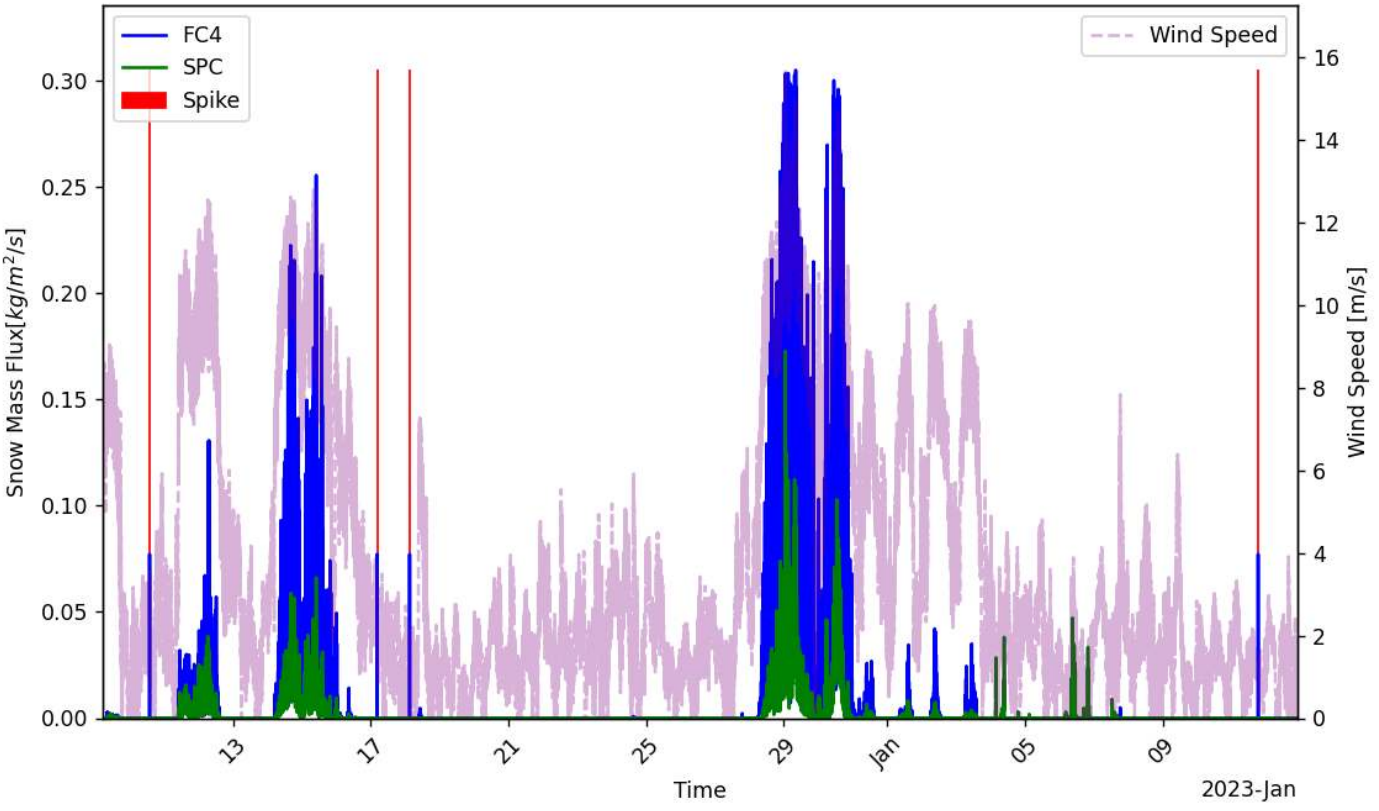


Figure 7: Zoomed view of a spike event with wind data shown for reference.

3.3 Snow surface elevation

The snow surface elevation measured by the SR50A sensor was initially filtered by removing all data points whose quality flags lay outside the 152 to 210 range. Two different methods were then applied

to further clean the dataset. The first method calculated the median absolute deviation (MAD) over a one-day window and removed all data points exceeding three times the MAD, followed by a one-day moving average on the remaining points. The second approach simply applied a one-day moving median. Although the first method was more complex, the moving median performed equally well, and even better under rapidly changing conditions, making it the preferred method.

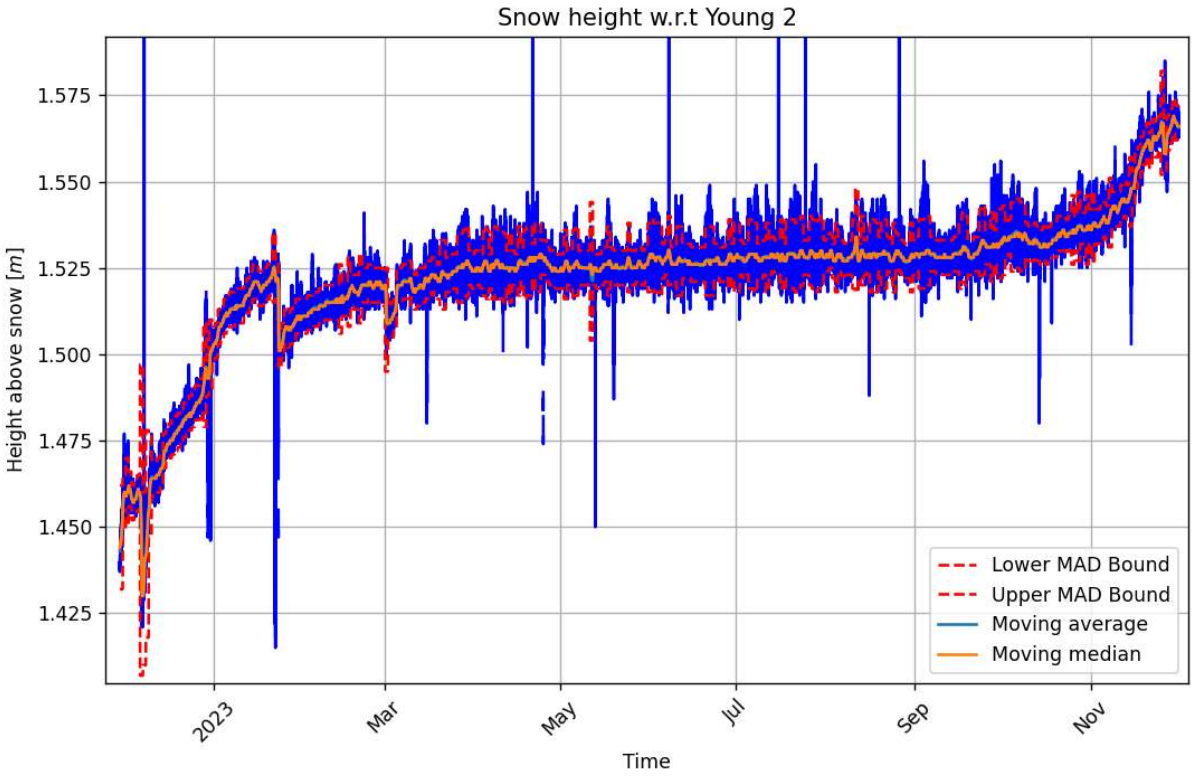


Figure 8: Snow surface elevation w.r.t the wind speed sensor over the entire measurement period.

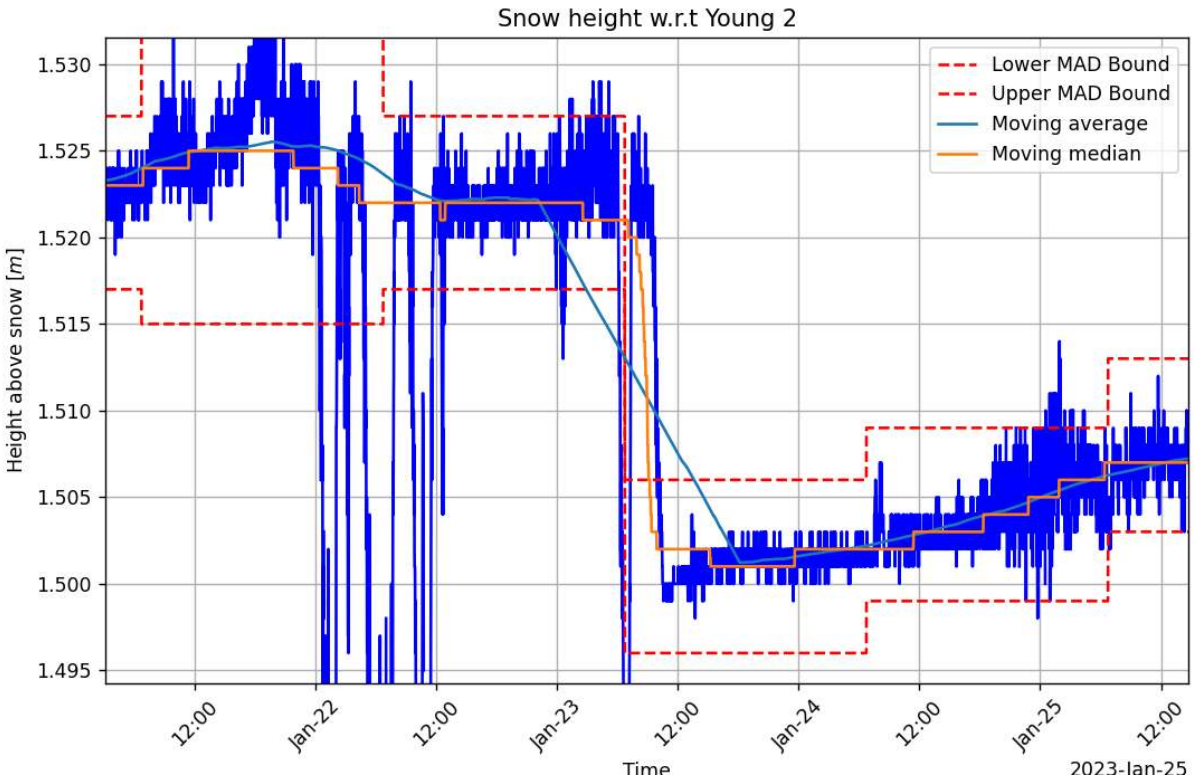


Figure 9: Snow surface elevation w.r.t the wind speed sensor during a period of rapid change.

3.4 Anemometer data

The fast anemometer data, sampled at 10 Hz, was processed using a provided script. The processing followed the method described in Sigmund et al. 2022, which includes the filtering of fast-response data using modified median absolute deviation (MAD) algorithms to detect and remove spikes.

4 Latent heat flux estimation methods

4.1 Eddy-covariance

The Eddy-covariance (EC) method enables a direct quantification of vertical turbulent fluxes by means of highfrequency measurements of the respective covariance components. The latent heat flux is given by

$$Q_L = L_w \, M_w \, \overline{w'c'} \tag{1}$$

Where Q_L ($\mathrm{W\,m^{-2}}$) is the latent heat flux, respectively, w' ($\mathrm{m\,s^{-1}}$), and c' ($\mathrm{mmol\,m^{-3}}$) are fluctuations of vertical wind speed and water vapor concentration, respectively, i.e., the deviations from the arithmetic mean of the averaging interval (10 min). The overbar denotes temporal averaging, M_w ($\mathrm{kg\,mol^{-1}}$) is molar mass of water, and L_w ($\mathrm{J\,kg^{-1}}$) is latent heat of evaporation or sublimation.

A positive Q_L represents sublimation, while a negative Q_L represents deposition. In principle, the Eddy-covariance method can capture the sum of surface and blowing snow sublimation that occurs below the measurement height.

4.2 Monin-Obukhov

According to the Monin-Obukhov similarity theory, the latent heat flux (W m^{-2}) can be expressed as (Monin and Obukhov 1954, King et al. 2001)

$$Q_L = -\rho L_w C_q \bar{u}(\bar{q}_z - \bar{q}_0), \tag{2}$$

where $\bar{q}_z$ and $\bar{q}_0$ (kg kg^{-1}) are average specific humidity at height z (m) and at the surface, respectively, $\bar{u}$ (m s^{-1}) is average wind speed at height z, and ρ (kg m^{-3}) is air density. The specific humidity at the surface was calculated from the measured surface temperature assuming saturation. The dimensionless exchange coefficient for moisture, C_q, is given by

$$C_q = \frac{\kappa^2}{\left[\ln\left(\frac{z}{z_{0q}}\right) - \Psi_q\left(\frac{z}{L}\right)\right]\left[\ln\left(\frac{z}{z_0}\right) - \Psi_m\left(\frac{z}{L}\right)\right]}, \tag{3}$$

where $\kappa = 0.4$ is the von Kármán constant, z_{0q} and z_0 (m) are the roughness lengths for humidity and momentum, respectively, Ψ_q and Ψ_m are the stability corrections for latent heat and momentum, respectively, z (m) is the sampling height, and L (m) is the Obukhov length. The momentum and sensible heat fluxes are parameterized using analogue equations (King, Anderson, and Smith 2001) .

The Monin-Obukhov (MO) bulk approach assumes that the phase change of water occurs at the surface and does not explicitly account for elevated water vapor sources or sinks caused by blowing snow particles. Thus, this approach is expected to underestimate Q_L in absolute magnitude during blowing snow events.

5 Latent heat flux computation

5.1 Eddy-covariance

The pre-processed fast anemometer data were then transferred to EddyPro, which applies multiple corrections to compute vertical turbulent fluxes of latent heat (water vapor), sensible heat (temperature), and momentum (shear stress). These corrections include adjustments for wind speed measurement offsets, tilt correction through coordinate rotation, time lag compensation, spectral corrections for high-pass and low-pass filtering effects, and density fluctuation adjustments. This data is then sent to another provided script, which computes quality flags for each data point. These flags indicate the reliability of the data and are defined in Sigmund et al. 2022 as follows:

- **Flag 0 (Good quality):** Data points with a quality flag of 0 meet all the criteria for high reliability. These points exhibit minimal missing data, steady-state fluxes, well-developed turbulence, and low levels of spikes. They are considered accurate and suitable for further analysis.
- **Flag 1 (Intermediate quality):** Data points with a quality flag of 1 partially meet the quality criteria. They may have moderate amounts of missing data (e.g., 10–25%), slight deviations from steady-state conditions or well-developed turbulence, or a small number of spikes. While these points are usable, caution is advised when interpreting them.
- **Flag 2 (Bad quality):** Data points with a quality flag of 2 fail to meet the quality criteria. They may have significant missing data (e.g., >25%), large deviations from steady-state, poorly

developed turbulence, or excessive spikes. These points are generally unreliable and should be excluded from further analysis.

The script evaluates various aspects of the flux data to determine these quality flags:

- **Missing data assessment:** The fraction of missing data for each flux (e.g., latent heat, sensible heat, momentum) is calculated. Missing data are flagged according to thresholds.
- **Steady-state test:** A test is conducted to check if the turbulent fluxes remain stationary over the averaging period. Fluxes that deviate significantly from steady-state conditions are flagged.
- **Well-developed turbulence test:** This test evaluates whether the turbulence characteristics meet theoretical expectations. Poorly developed turbulence is flagged.
- **Spike analysis:** The percentage of spikes in high-frequency data (e.g., wind components, water vapor) is computed. Excessive spikes are removed and produce data gaps which leads to a lower quality flag for the affected fluxes.
- **Combined flagging:** Individual flags (e.g., for missing data, steady-state test, turbulence) are combined into an overall quality flag for each flux, accounting for interdependencies between fluxes.

5.2 Monin-Obukhov

A provided script was used to compute the latent heat flux. This script requires several input parameters, including measurement height, wind speed, humidity, temperature, roughness length (for flat homogeneous surface like snow, the roughness lengths for momentum and humidity can be assumed to be equal ($z_{0q} = z_0$)), and pressure. All those parameters were available from the measurements, except for the roughness length, which was estimated, as described in Roughness length estimation.

All input parameters were provided to the script's iterative algorithm, which refines the latent heat flux calculation until convergence is reached. Since different stability correction functions have been proposed in the literature, the following functions were selected: the method described in Holtslag and De Bruin 1988 for stable and neutral stratification, the approach outlined in Paulson 1970 for wind speed in unstable conditions, and the formulation presented in Stearns and Weidner 1993 for temperature and humidity in unstable conditions.

5.2.1 Roughness length estimation

Since the roughness length was not directly available an estimation was computed. Under near-neutral stratification conditions ($-0.1 < z/L < 0.1$), the wind profile is assumed to follow a logarithmic distribution. Consequently, z_0 was calculated for these periods without the need for additional stability corrections. Under these conditions, the roughness length was defined as:

$$z_0 = z \cdot \exp\left(-\kappa \frac{\overline{U}}{u_*}\right), \tag{4}$$

where z is the measurement height (m), $\kappa = 0.4$ is the von Kármán constant, $\overline{U}$ is the average wind speed at height z (m s^{-1}), and u_* is the friction velocity (m s^{-1}) obtained from EddyPro outputs. The Obukhov length L provides a measure of stability in the surface layer. Direct roughness length

estimates from raw data are often noisy and have gaps when conditions don't satisfy the near-neutral stratification threshold. Since z_0 is expected to vary slowly, a rolling median filter over a four-week window was applied to smooth the estimates. This approach yields a more stable and representative time series of roughness length values.

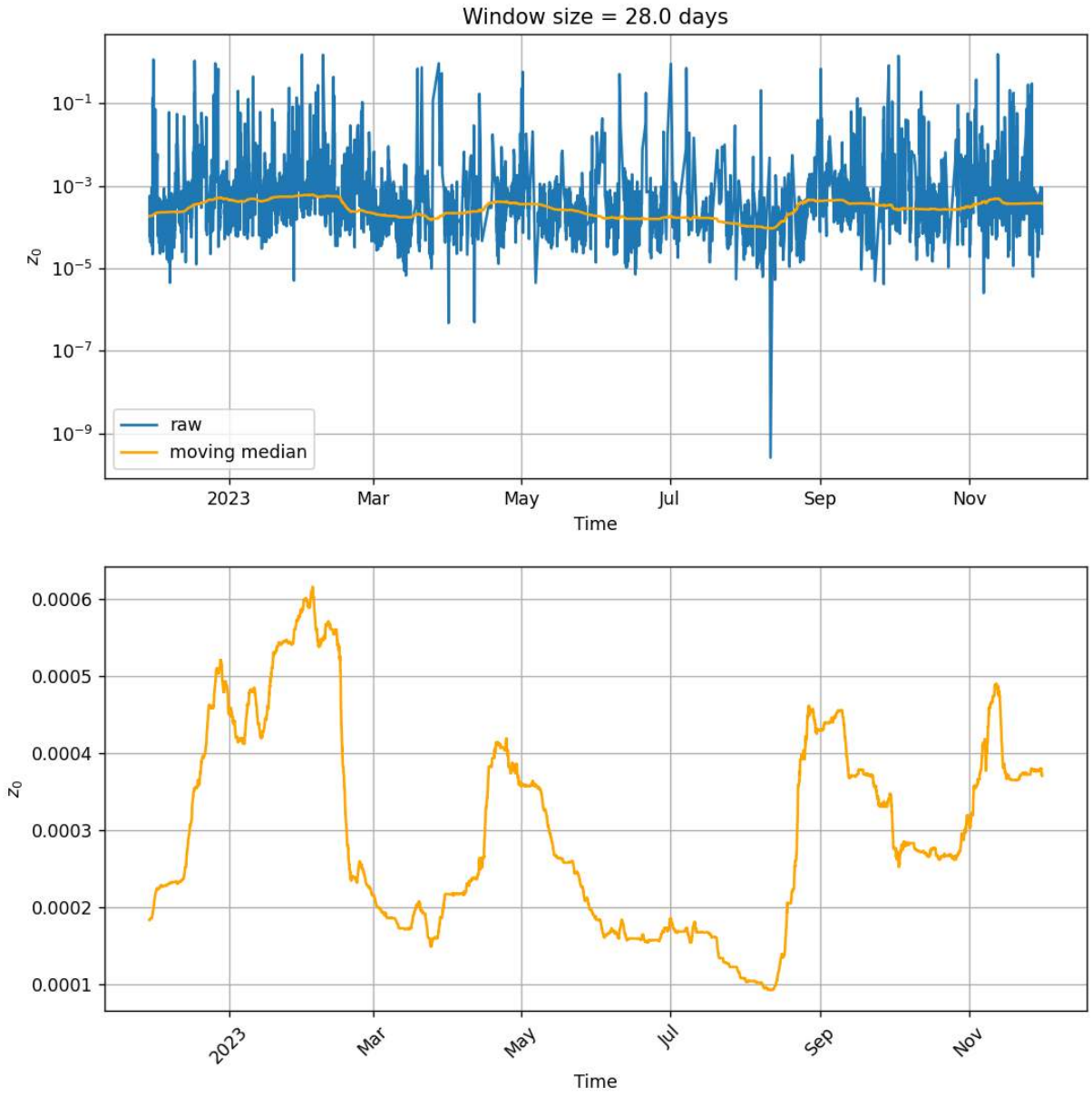


Figure 10: Roughness length estimates over time after applying a four-week rolling median.

6 Analysis

6.1 Snow transport events

The objective of this study is to quantify latent heat fluxes under different snow transport conditions. According to Trouvilliez et al. 2015, the FC4 sensor has a detection threshold of $0.001\,\mathrm{g\,m^{-2}\,s^{-1}}$. Consequently, all periods during which the snow flux was below this threshold were classified as non blowing

snow events. Based on the remaining data, quantiles were computed to categorize snow events into 5 classes.

Quantile	**Snow Flux** ($\mathrm{kg\,m^{-2}\,s^{-1}}$)
Q1	3.14×10^{-5}
Q2	3.74×10^{-4}
Q3	2.8×10^{-3}
Q4	2.17×10^{-2}

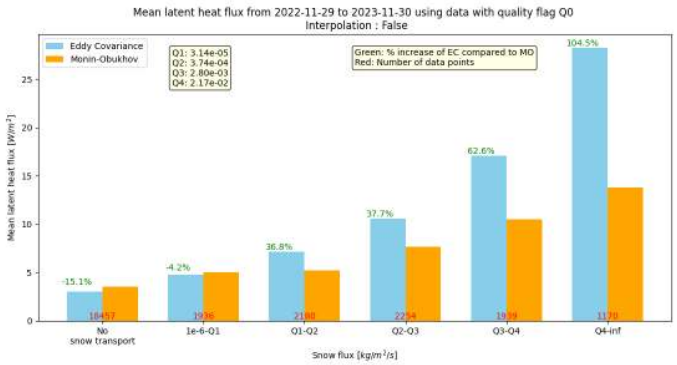
Table 3: Quantiles of snow flux ($\mathrm{kg\,m^{-2}\,s^{-1}}$) after filtering by the threshold.

These quantiles in table 3 illustrate the distribution of snow flux values across several orders of magnitude, enabling a study of latent heat fluxes associated with each class.

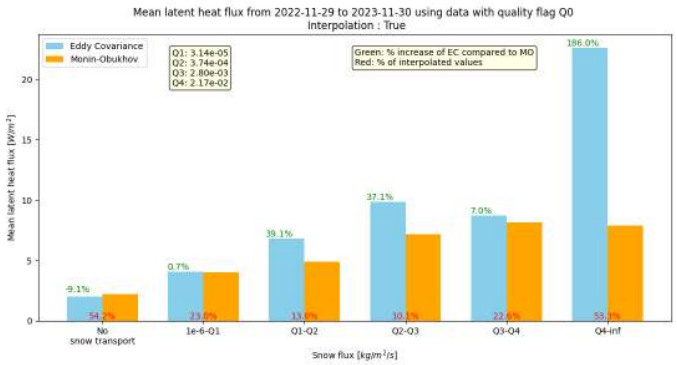
6.2 Mean latent fluxes

The mean latent heat fluxes over the year were computed using both methods and four slightly different datasets. The inclusion of Eddy-covariance latent heat flux data was adjusted based on their respective quality flags. Calculations were performed using only data with a quality flag of 0 (hereafter referred to as Q0), considered the most reliable, as well as data with quality flags of 0 and 1 (hereafter referred to as Q01). Data with a quality flag of 2 were deemed unreliable and therefore excluded from all computations. Whenever Eddy-covariance data were omitted, the corresponding time periods were also excluded from the Monin-Obukhov flux calculations to maintain consistency. Secondly, the impact of handling missing values was examined, specifically whether they should be interpolated linearly or omitted from the calculation of the annual mean flux. In both approaches, the resulting mean fluxes were evaluated for each distinct snow transport intensity class.

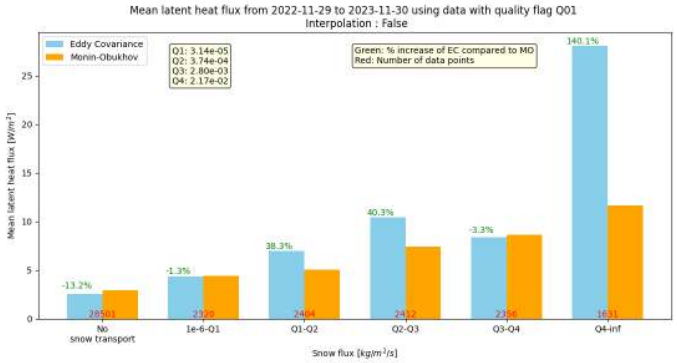
The results of these methodological variations are illustrated in Figure 11.



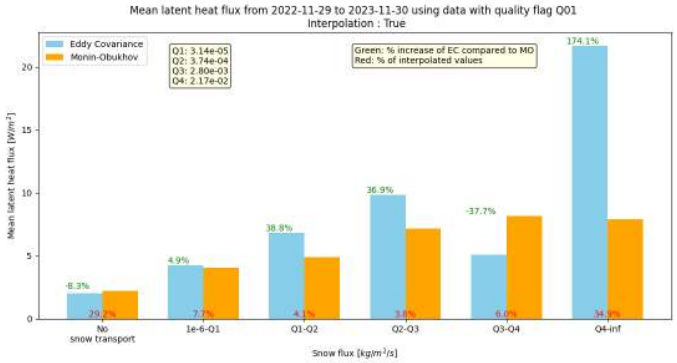
(a) Q0 (non-interpolated)



(b) Q0 (interpolated)



(c) Q01 (non-interpolated)



(d) Q01 (interpolated)

Figure 11: Comparison of the mean latent heat fluxes

Since the air in Antarctica is very dry, it is expected that sublimation occurs more frequently than deposition, and therefore the mean latent heat flux will be positive. The results confirm that the mean latent heat flux increases with the intensity of snow transport events. This increase is more clearly captured by the Eddy-covariance method, whereas the Monin-Obukhov method fails to capture the full magnitude of the flux.

For all four datasets, the mean latent heat flux calculated using the eddy-covariance method generally increases with snow transport intensity. Although a similar trend appears when using the Monin-Obukhov method, the increase is significantly weaker. In addition, the percentage difference (w.r.t MO) between the eddy-covariance and Monin-Obukhov methods grows with snow transport intensity, highlighting the limitations of the Monin-Obukhov approach in representing fluxes associated with blowing snow.

While a general increase in flux and in the percentage difference between eddy-covariance and Monin-Obukhov methods is observed across all four datasets, only $Q0$ (non-interpolated) exhibits a strictly monotonic rise. Specifically, in the snow transport class $Q3$–$Q4$, the mean latex flux is lower when interpolating or including lower quality data points, which suggests that deposition may be estimated differently in those datasets. Additionally, it is notable that the highest snow transport class contains the largest proportion of missing values. This is likely because snow particles may interfere with the anemometer sensors, introducing measurement artifacts.

Over the course of the year, snow transport occurred during 23.7% of the time. The annual mean latent heat fluxes for periods with and without snow events are compared across the different datasets in Table 4.

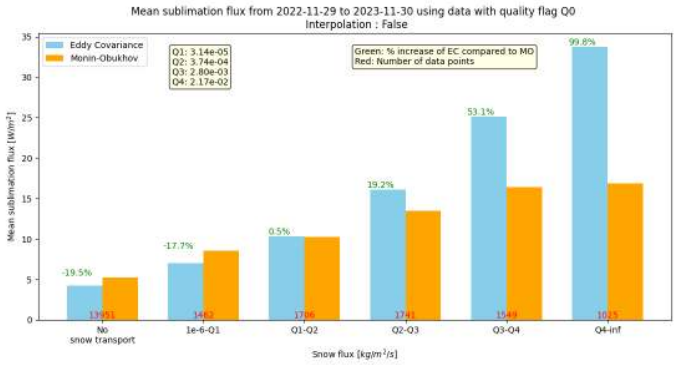
Method	Dataset	Q0		Q01	
		Interpolated	Raw	Interpolated	Raw
EC	Snow event	10.4	12.1	9.53	10.6
	No snow event	2.02	3.02	2.04	2.55
MO	Snow event	6.44	7.9	6.44	7.17
	No snow event	2.22	3.55	2.22	2.93

Table 4: Mean latent heat fluxes in $[W/m^2]$ over the whole year for the EC and MO methods, with different datasets, with and without snow transport.

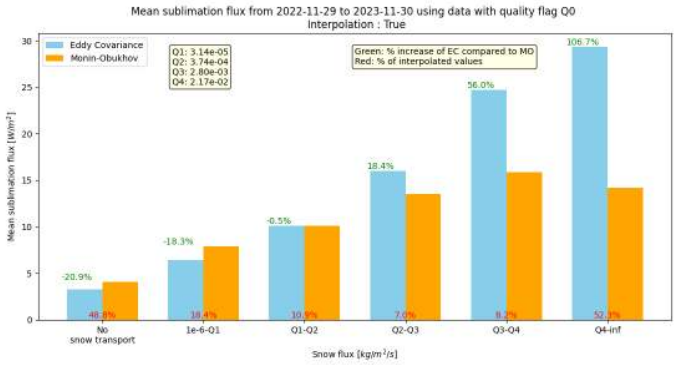
These results confirm that the MO method cannot be reliably used during periods of snow transport.

6.3 Differences between sublimation and deposition

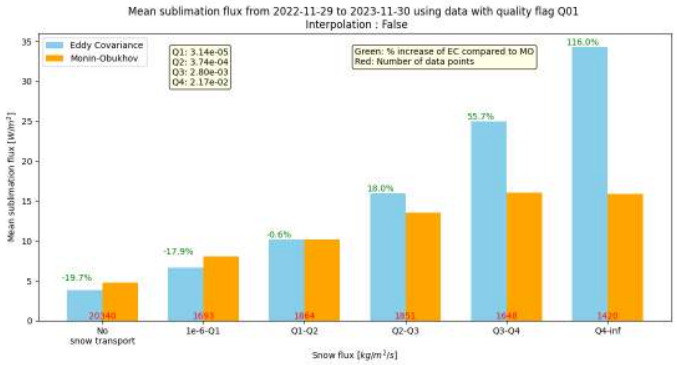
To further investigate the contributions to the latent heat flux, the total flux was separated into sublimation and deposition components (separately for each method). Figures 12 and 13 present the results for sublimation and deposition, respectively.



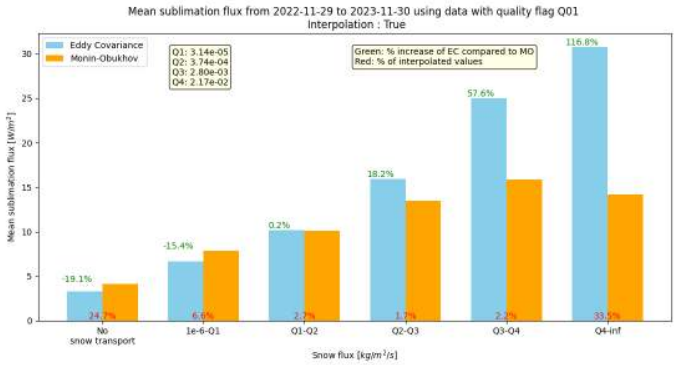
(a) Q0 (non-interpolated)



(b) Q0 (interpolated)



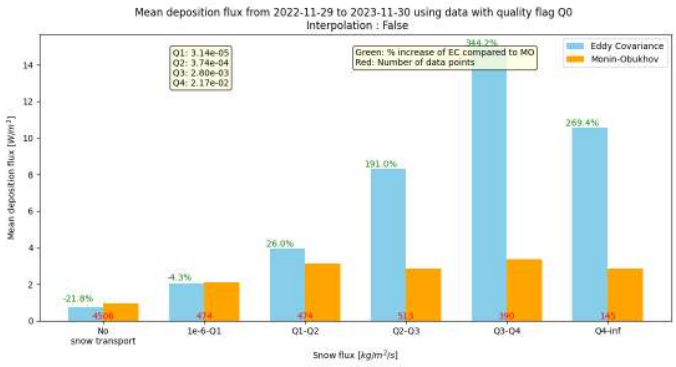
(c) Q01 (non-interpolated)



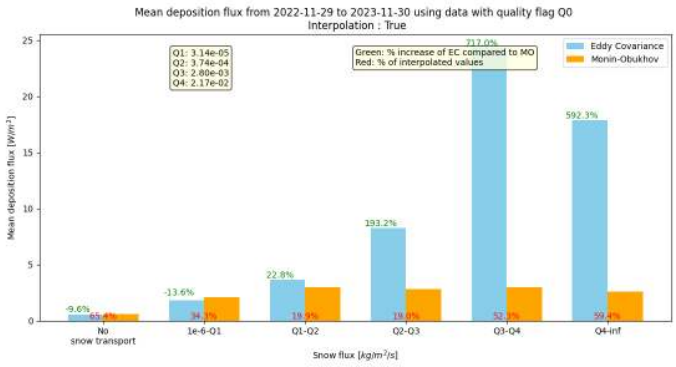
(d) Q01 (interpolated)

Figure 12: Comparison of the mean sublimation fluxes for all datasets.

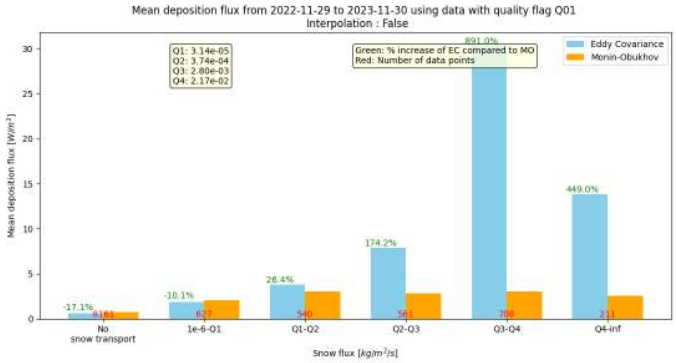
As shown in Figure 12, sublimation events follow the expected trend: stronger snow transport correlates with higher sublimation fluxes. This pattern is evident in both the magnitude of the fluxes and in the relative differences between eddy-covariance and Monin-Obukhov estimates, and it holds true for all four datasets.



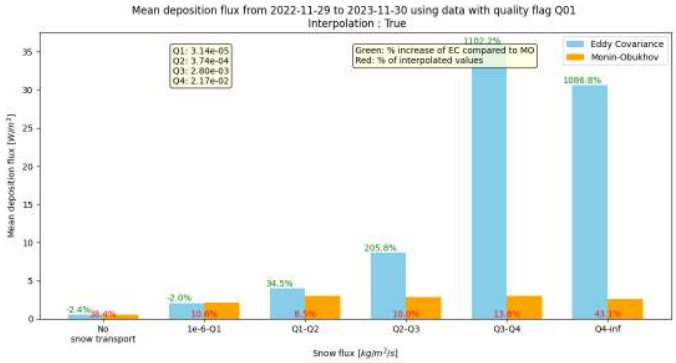
(a) Q0 (non-interpolated)



(b) Q0 (interpolated)



(c) Q01 (non-interpolated)



(d) Q01 (interpolated)

Figure 13: Comparison of the mean deposition fluxes for all datasets.

In contrast, Figure 13 shows that deposition does not peak at the strongest snow transport classes but rather in an intermediate class (Q3–Q4). Notably, in the Q0 non-interpolated dataset, the peak affects less the mean latent heat flux because its class contains only 390 valid data points (with a quality flag of 0), compared to 708 when including data points flagged as 1. Therefore, when data with quality flag 1 are included or when data is interpolated, the mean latent heat flux is more strongly affected by this deposition peak.

There could be two hypothesis for this behavior:

1. The strongest snow transport class is significantly under-sampled during deposition events. Specifically, this class contains only 145 data points on average compared to 462 for other snow transport classes with quality flag 0, and 211 data points compared to an average of 609 for quality flag 01.

   This under-sampling effect likely arises from the fundamental relationship between atmospheric humidity and deposition. In Antarctica, when large masses of humid air move in from the ocean, the result is often strong winds correlated with snow transport. These strong winds, combined with the high moisture content, lead to heavier snowfall. In essence, the more humid the incoming air, the greater the deposition—and consequently, the more intense the snowstorms. Snowfall can negatively impact sensor performance, leading to the removal of affected data points during preprocessing. Since these events can produce large gaps, exclusion of these periods as well as interpolation could totally ignore the real magnitude of the deposition. Consequently, deposition events under very strong snow transport conditions are not measured reliably.

2. Conditions favorable for deposition (air near saturation) might occur more frequently at moderate wind speeds rather than at the strongest winds, where turbulence may reduce local humidity.

7 Conclusion

This study confirms that the Monin-Obukhov method is unsuitable for quantifying latent heat fluxes during snow transport events, whereas the Eddy-covariance approach provides more accurate flux measurements. Nonetheless, the results suggest that deposition events during intense snow transport may be underrepresented in the EC dataset, potentially due to sensor interference under heavy snowfall conditions.

A promising avenue of future research involves a more thorough examination of snowfall to determine if it coincides with the identified data gaps. If snowfall is indeed responsible for these gaps, it would be useful to develop a model that does not rely on data severely impacted by blowing snow, such as the high-frequency anemometer measurements, and would incorporate variables that capture horizontal snow transport. By fitting this model to the reliable EC flux measurements, its performance could be evaluated with respect to accurately reproducing the observed latent heat fluxes. A sufficiently robust model might then be used to fill missing EC data, thereby improving the completeness and reliability of flux estimates during snowfall events. Such an approach would help to address current limitations in capturing deposition phenomena under intense blowing snow, ultimately enhancing our understanding of surface mass and energy balances in polar regions.

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